

requirements of the other applications. One of the main tasks of a tier-ing initiative is to classify the data created based on several attributes such as age of the data, its business value, criticality and its usage.

The value of the information created in businesses changes with time. This is not to mean that it isn't useful. The emergence of regulatory compliance laws such as the Sarbanes Oxley (SOX) and other corporate best practices has necessitated the need for archiving old data from organisational e-mails, sales data, and so on, for a specified period of time. Though this is a low priority segment of data and has a low frequency of usage, it nevertheless needs to be handled in a specified manner, with the right archival policies.

Business-critical data requires a high priority and an equally robust mechanism to ensure the reliability and availability of such data. This type of data includes sales reporting data, IP, clinical research data, and so on, which demand high priority in terms of data back-up, restoration, etc.

Every organisation creates different types of data to support its business. There is primary data, which is critical to the business, whereas the other forms are purely for business support. Some of the secondary and tertiary types of data are created through tools such as BI applications for business analysis. Hence, it is important to categorise the information generated based on the usage of the information for different applications within the organisation.

Case Study - Implementation of tiering

In the case of a particular healthcare company, the data centre was consolidated on a single premium vendor's storage platform. Due to market conditions and increasing pressure to cut IT budgets, the organisation was forced to look at ways to cut excess expenditure on operational costs. This drove the focus towards the business justification of expending premium storage for all applications, irrespective of their requirements.

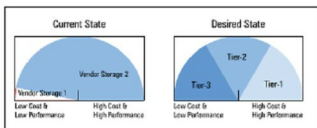


Figure 2: Storage vs applications

Hence, the management started to take care of new storage requirements on cheaper storage to balance the operational costs on the premier storage.

In spite of the presence of an adequate storage infrastructure, the company faced severe bandwidth issues for about seven of the 45 critical business applications. This was followed by lower utilisation levels on the storage, which was at an unusual level of 45 per cent. The existence of excess capacity as a result of application owners incorporating a 10-30 per cent buffer in their storage requirements over several years, contradicted the fact that the business applications faced availability issues.

This baffled the management and clearly pointed to the fact that there was no clear policy and methodology in place for treating applications' requirements on the merit of their functional and non-functional requirements. The potential danger of risking downtime and data loss on business-critical applications prompted the management to take up a portfolio analysis, with the ultimate goal of creating a data life cycle management model for all the business applications through a tiered infrastructure [refer to Figure 2].

A data heat map traces the various phases of the data's life cycle and the business value that it provides at various stages. This is an important tool to build an effective tiered infrastructure model, as it provides valuable information of the functional and non-functional requirements of the application data at various

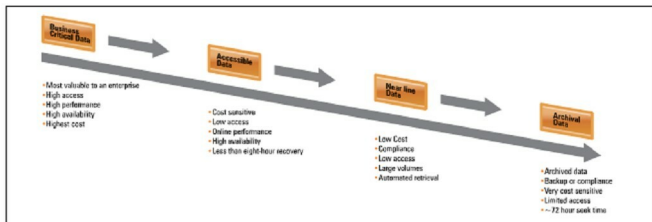


Figure 3: Data life cycle management model for all the business applications through a tiered infrastructure